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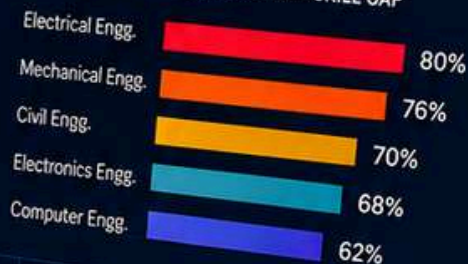
WHICH ENGINEERING BRANCHES FACE THE BIGGEST SKILL GAP?

Engineering education is evolving, but industry expectations are changing faster. Which branches are industry-ready, and which are falling behind? Let's find out with data.

SKILL GAP OVERVIEW



TOP 5 BRANCHES WITH HIGHEST SKILL GAP



INDUSTRY DEMAND VS CURRICULUM COVERAGE



SKILL CATEGORY GAP ANALYSIS



PROJECT TITLE

BRANCH-WISE SKILL GAP & INDUSTRY READINESS INTELLIGENCE PLATFORM



Identify branch-wise employability gaps, compare curriculum with industry demand, analyze recruiter expectations, build branch-specific skill intelligence models, and design upskilling recommendations.



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PREPARED BY



Koushik Garg
Data & Business Analyst



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UNDER THE GUIDANCE OF



Wasim Patwari
Mentor & Guide



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Data-Driven Skill Gap Identification



Branch-wise Industry Readiness Scoring



Recruiter Demand & Expectation Analysis



Curriculum vs Industry Alignment Insights



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www.analyticscareerconnect.com



info@analyticscareerconnect.com



+91 9607157409

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Executive Summary

This report analyzes branch-wise employability and skill-gap patterns across 2,813 engineering colleges in India, covering 28,091 branch-level records across seven disciplines: Computer Science Engineering (CSE), Information Technology (IT), Electronics & Communication Engineering (ECE), Electrical Engineering (EEE), Mechanical Engineering, Civil Engineering, and Emerging Technology Domains. The objective is to identify which branches face the largest gap between the skills colleges teach and the skills the industry demands, and to translate that gap into actionable, branch-specific upskilling recommendations for ACC.

The analysis converts placement outcomes, salary data, internship participation, recruiter demand, and a computed Skill Readiness Index into a single comparable Skill Gap Score for every branch. The findings show a clear and consistent divide: traditional core branches — Civil Engineering and Mechanical Engineering — carry the largest skill gaps and the weakest placement outcomes, while technology-facing branches — CSE, Emerging Technology, and IT — are the most industry-ready.

Key Findings at a Glance

- **Widest skill gap:** Civil Engineering, with a Skill Gap Score of 52.8/100 and the lowest placement rate (49.6%) and lowest average salary (₹4.32 LPA) of any branch.
- **Second widest:** Mechanical Engineering, Skill Gap Score 46.8/100, placement rate 54.0%.
- **Most industry-ready:** Computer Science Engineering, Skill Gap Score 27.8/100, placement rate 74.6%, average salary ₹7.21 LPA — the benchmark other branches are measured against.
- **Consistent pattern:** The gap between what is taught and what recruiters demand is largest in branches where curricula still emphasize manual/classical methods (2D drafting, manual surveying, theory-only circuit design) rather than the modern software and digital tools recruiters now expect.
- **Government vs. Private divide:** Within every branch, government institutions outperform private ones on placement by 10–14 percentage points, with Civil Engineering showing the widest gap (13.8 points).
- **Recommendation priority:** Civil and Mechanical Engineering require immediate, structured upskilling interventions; ECE and EEE need moderate, targeted digital-tools intervention; CSE, IT and Emerging Technology need continuous upskilling to keep pace with fast-moving demand (GenAI, cloud-native, MLOps) rather than remedial intervention.

Project Background

India produces millions of engineering graduates every year across CSE, IT, ECE, EEE, Mechanical, Civil, and emerging technology domains. Despite obtaining engineering degrees, a significant share of graduates struggle to secure employment because the skills acquired in college do not consistently align with industry expectations.

This skill gap is not uniform — it varies sharply by branch. Some students lack exposure to modern technical tools; others lack the communication, analytical thinking, problem-solving, project exposure, or industry awareness that recruiters look for. The DATAYUG project was commissioned to identify these branch-specific employability gaps and provide data-driven recommendations that ACC can act on.

2.1 Business Problem

ACC wants this project to answer five questions:

- Which engineering branches face the largest skill gaps?
- Which skills are missing in each branch?
- What is the gap between curriculum and industry expectations?
- Which branches require immediate upskilling support?
- How can branch-specific learning pathways improve employability?

2.2 Project Objectives

- Identify branch-wise employability gaps.
- Compare academic curriculum with industry demand.
- Analyze recruiter expectations.
- Build branch-specific skill intelligence models.
- Design data-driven upskilling recommendations.

Methodology & Data Overview

3.1 Data Source

The analysis draws on a single consolidated branch-intelligence database (FINAL_DATA.xlsx) containing 28,091 college-branch records spanning 2,813 colleges across 483 locations (states/cities) and 30 institutional categories (IITs, NITs, IIITs, state government colleges, private universities, and more). Each record captures placement percentage, average salary, internship participation, curriculum subjects, recruiter demand narrative, industry-required skills, top recruiters, skills taught vs. skills demanded, the explicit missing-skills gap, emerging technology requirements, and an employability indicator.

3.2 Skill Gap Score — Methodology

For every branch, a Skill Readiness Index (0–100) was computed from placement outcomes, recruiter demand intensity, and internship/industry exposure. The Skill Gap Score used throughout this report is defined as:

$$\text{Skill Gap Score} = 100 - \text{Skill Readiness Index}$$

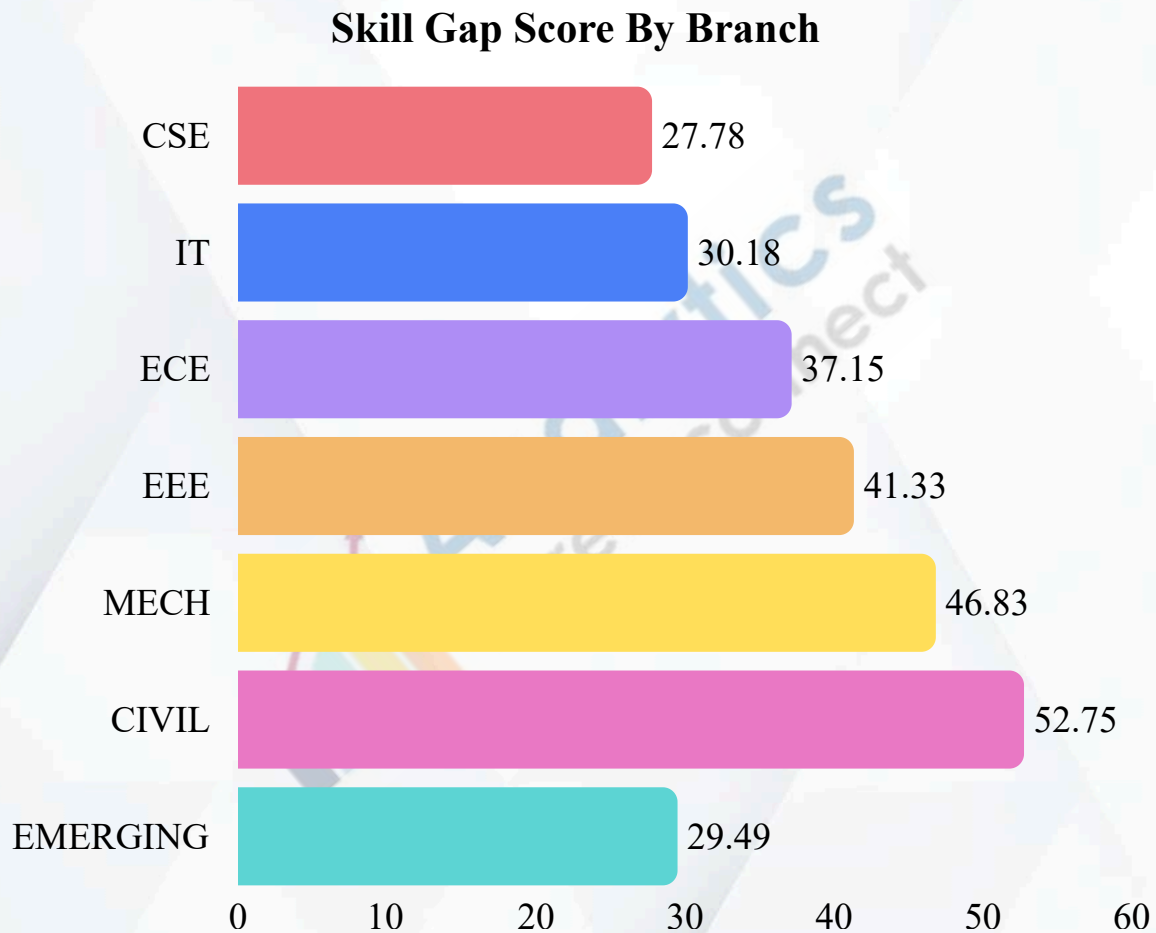
A higher Skill Gap Score indicates a larger shortfall between what a branch's graduates currently offer and what the industry expects — and therefore a higher priority for intervention.

3.3 Scope

- Branches covered: CSE, IT, ECE, EEE, Mechanical, Civil, Emerging Technology Domains.
- Comparison dimensions: Placement rate, average salary, internship participation, recruiter demand, curriculum-vs-industry skill match, and ownership type (Government vs. Private).
- Every branch is represented by 2,813 college-level records, i.e., one record per college × branch combination, making the comparison statistically balanced across branches.

Branch-wise Skill Gap Analysis

The chart below ranks all seven branches by Skill Gap Score, from the widest gap (top) to the narrowest (bottom). Civil Engineering and Mechanical Engineering sit well above the rest, confirming that core, infrastructure-oriented branches face the most urgent industry-readiness challenge.



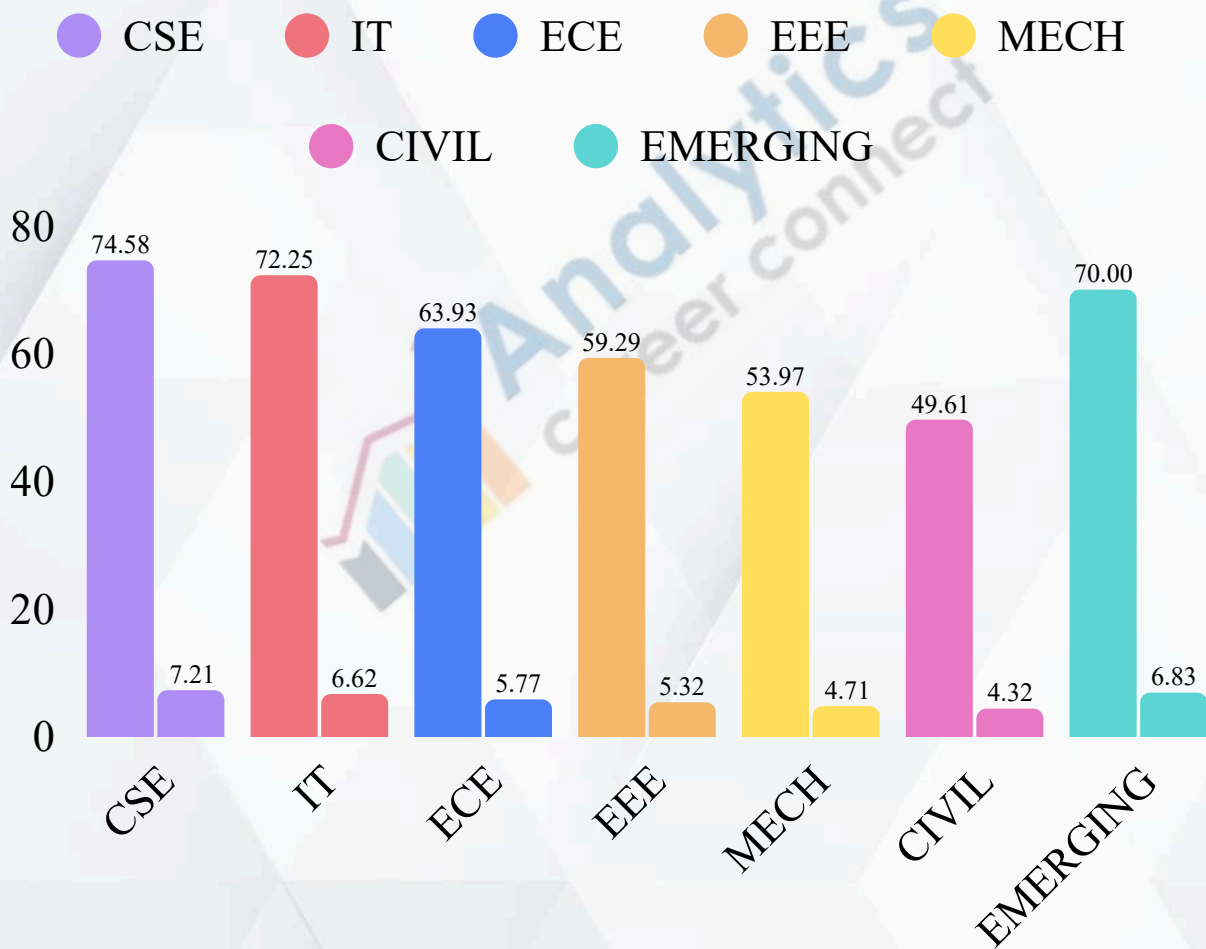
Branch	Placement %	Avg. Salary (LPA)	Internship %	Skill Readiness	Skill Gap Score	Priority
Civil	496	₹4.32	416	473	528	Critical
MECH	540	₹4.71	461	532	468	High
EEE	593	₹5.32	518	587	413	High
ECE	639	₹5.77	561	629	371	Moderate
IT	723	₹6.62	625	698	302	Low
EMERGI NG	700	₹6.83	615	705	295	Low
CSE	746	₹7.21	648	722	278	Low

This table consolidates the five core employability metrics — placement percentage, average salary, internship participation, Skill Readiness Index, and the derived Skill Gap Score — into a single side-by-side view for all seven branches. Reading down the Priority column shows the four-tier urgency scale used throughout this report: Critical (Civil), High (Mechanical, EEE), Moderate (ECE), and Low (IT, Emerging Technology, CSE). The table confirms that every metric moves in the same direction together — the branches with the lowest placement percentage also have the lowest salary, lowest internship participation, and lowest skill readiness, with no branch bucking the trend on any single column. Civil Engineering anchors the bottom of every metric simultaneously, while CSE anchors the top of every metric simultaneously, which is what makes the Skill Gap Score a reliable single summary number rather than an average that hides disagreement between metrics. This table is the reference point every other table and chart in the report builds on or drills into further.

Placement Percentage vs. Average Salary

Placement percentage and average salary move together almost perfectly across branches — branches with the weakest placement outcomes also command the lowest starting salaries, and vice versa. This confirms that the skill gap directly translates into an employability and earnings penalty, not just a hiring-rate issue.

Placement % Vs Avg Salary By Branch



Curriculum vs. Industry Expectation Gap

For each branch, the table below contrasts what colleges currently teach with what recruiters actively demand, and lists the explicit missing-skills gap and emerging technology requirements captured in the database. Branches are presented in descending order of urgency (Skill Gap Score).

5.1 Civil Engineering — Skill Gap Score: 52.8 (Critical)

Dimension	Details
Skills currently taught	Basic 2D CAD Drafting, Manual Surveying (Chain/Theodolite), Concrete Mix Design Theory, Manual Structural Calculations
Skills recruiters demand	BIM (Revit), STAAD Pro/ETABS Structural Software, Total Station & GIS, Primavera P6 Project Scheduling
Missing skills gap	BIM 3D/4D Integration, Advanced Structural Software (ETABS), Digital Project Management (Primavera/MS Project), Drone Surveying
Emerging tech requirements	Smart Infrastructure & IoT Sensors, Pre-fabricated Construction, Green Building Certification (LEED/GRIHA), 3D Concrete Printing
Recruiter demand	Moderate — driven by the Smart Cities Mission, NHAI highways, Railways, Metro Rail, and real-estate development.
Representative top recruiters	L&T, Tata Projects, Shapoorji Pallonji, NHAI, DLF, Gammon India, AFCONS, RITES

This table shows why Civil Engineering carries the widest skill gap of any branch in the database. The 'Skills currently taught' row is dominated by manual and 2D methods — chain/theodolite surveying and basic CAD drafting — while the 'Skills recruiters demand' row is entirely software-driven, centered on BIM, ETABS, and digital project scheduling. The four-item missing-skills list is essentially the industry-demand row restated as a gap, which is why the Skill Gap Score for this branch (52.8) is the highest recorded. Recruiter demand is only 'Moderate' despite large government infrastructure programs, because the roles that exist are increasingly filtered through digital-tool proficiency that current graduates lack. The recruiter list (L&T, NHAI, DLF, and similar) confirms the demand exists — the bottleneck is graduate readiness, not job availability.

5.2 Mechanical Engineering — Skill Gap Score: 46.8 (High)

Dimension	Details
Skills currently taught	2D Drafting (AutoCAD), Basic Machine Shop Workshop, Classical Thermodynamics Math, Manufacturing Processes Theory
Skills recruiters demand	3D Parametric Modeling (SolidWorks/CATIA), Finite Element Analysis (ANSYS), Computational Fluid Dynamics (CFD), GD&T, Mechatronics/Robotics
Missing skills gap	Advanced FEA/CFD Simulation, GD&T Tolerancing, Automated CNC Programming, EV Vehicle Dynamics
Emerging tech requirements	Additive Manufacturing/3D Printing, Autonomous Robotics, Digital Twin Modeling, Industry 4.0 IoT
Recruiter demand	Moderate to High — strong pull from Automotive, EV manufacturing, Aerospace, Defense and Heavy Industries.
Representative top recruiters	Tata Motors, Maruti Suzuki, Mahindra & Mahindra, L&T, BHEL, ISRO, Boeing, Bosch

This table places Mechanical Engineering second-highest on the urgency scale (46.8), driven by a similar pattern to Civil: strong grounding in classical theory (thermodynamics, manufacturing processes) but limited exposure to the simulation and automation tools recruiters expect. The demand row spans 3D modeling, FEA/CFD simulation, and mechatronics — all skills that require dedicated lab and software access most curricula have not yet built in. The 'Moderate to High' recruiter demand reflects strong pull from automotive, EV manufacturing, and defense, sectors that are actively hiring but selecting for the specific missing skills listed. Automated CNC programming and EV vehicle dynamics stand out as the newest gaps, tracking the industry's shift toward electric and automated manufacturing rather than traditional internal-combustion engineering.

5.3 Electrical Engineering (EEE) — Skill Gap Score: 41.3 (High)

Dimension	Details
Skills currently taught	AC/DC Machine Theory, Power System Calculations, Basic Transformer Design, Relay Protection Theory, Control Systems Math
Skills recruiters demand	Power Electronics Converter Design, MATLAB/Simulink Modeling, EV Battery Management Systems (BMS), PLC/SCADA Automation, ETAP/PSCAD
Missing skills gap	EV Powertrain Design, Smart Battery Management Systems, Industrial SCADA Integration, Renewable Microgrid Design
Emerging tech requirements	EV Charging Infrastructure, Silicon Carbide (SiC) Power Devices, Smart Grid IoT, Solid-State Battery Systems
Recruiter demand	High — driven by EVs, the green-energy transition, smart grids and infrastructure projects.
Representative top recruiters	NTPC, PowerGrid, BHEL, Schneider Electric, ABB, Siemens, Tata Power, Ather Energy

This table shows EEE transitioning from a 'High' to 'Moderate' gap branch, with a skill mismatch centered on the shift from classical power-systems theory to power electronics and renewable/EV-linked applications. The taught row (AC/DC machine theory, transformer design) reflects a curriculum built for the grid infrastructure of a prior era, while the demanded row (EV battery management, SCADA automation, ETAP/PSCAD) reflects where hiring is actually concentrated today. Recruiter demand is explicitly 'High,' driven by the EV transition and smart-grid rollouts, and the recruiter list (NTPC, Tata Power, Ather Energy) spans both legacy utilities and new-energy companies — indicating EEE graduates who close this gap have access to both established and high-growth employers.

5.4 Electronics & Communication Engineering (ECE) — Skill Gap Score: 37.1 (Moderate)

Dimension	Details
Skills currently taught	Basic Analog Circuit Design, 8085/8086 Assembly, Basic Microcontroller Programming, MATLAB Simulation, Digital Logic Theory
Skills recruiters demand	Verilog/SystemVerilog, Cadence EDA Suite, RTL Design & Physical Verification, Embedded C/C++, ARM Cortex Architecture, RTOS, PCB Design
Missing skills gap	Advanced VLSI Physical Design, Modern EDA Tool Expertise, Firmware/RTOS Development, High-Speed PCB Layout
Emerging tech requirements	Semiconductor FinFET/GAA Architecture, 5G/6G RF Front-End Design, RISC-V Architecture, Automotive SoC
Recruiter demand	High and rapidly growing — fuelled by the India Semiconductor Mission, 5G/6G rollout and EV electronics.
Representative top recruiters	Qualcomm, Texas Instruments, Intel, NVIDIA, Samsung R&D, ISRO, Bosch

This table captures ECE's position just past the midpoint of the urgency scale (37.1, Moderate), with a curriculum still built around classical analog and assembly-level programming while recruiters have moved to modern chip-design and embedded workflows. The demand row (Verilog, Cadence EDA, ARM Cortex, RTOS) reflects the semiconductor and embedded-systems industry's actual toolchain, and the missing-skills row shows the gap is concentrated in VLSI physical design and firmware development specifically. Recruiter demand is described as 'High and rapidly growing,' tied directly to the India Semiconductor Mission and 5G/6G rollout referenced in this report's sources — meaning this branch's gap is the most likely to close quickly if curriculum investment follows the policy and hiring momentum already underway.

5.5 Information Technology (IT) — Skill Gap Score: 30.2 (Low)

Dimension	Details
Skills currently taught	C, Java, Relational Databases, Web Programming Basics, Software Testing Fundamentals, Network Protocols
Skills recruiters demand	Full-Stack Web Development (MERN/MEAN), Cloud Native Architecture, API Security, Automated Testing (Selenium/Cypress), Python Scripting, DevOps
Missing skills gap	Cloud Security Frameworks, Kubernetes Orchestration, Modern Frontend Frameworks, Production Code Quality
Emerging tech requirements	Zero-Trust Security, Cloud-Native Microservices, Serverless Computing, AI-Assisted Testing
Recruiter demand	Very High — broad demand across IT Services, Enterprise Solutions, FinTech and Product firms.
Representative top recruiters	TCS, Infosys, Cognizant, Wipro, Accenture, Oracle, Cisco, SAP

This table shows IT already operating in the 'Low' urgency band (30.2), but the structure of the gap is different from the core branches above: the taught row (C, Java, relational databases) is a reasonable foundation, and the demanded row (full-stack frameworks, cloud-native architecture, DevOps) is an extension of that foundation rather than a replacement for it. The missing-skills row — cloud security, Kubernetes, modern frontend frameworks — reflects the pace of change in web and cloud engineering rather than a structural mismatch. Recruiter demand is 'Very High' and broad-based across IT services, enterprise software, and FinTech, so the practical implication of this table is that IT graduates need continuous skill refreshes, not remedial catch-up.

5.6 Emerging Technology Domains — Skill Gap Score: 29.5 (Low)

Dimension	Details
Skills currently taught	Python Basics, Introductory Machine Learning Theory, Data Science Math/Statistics, Basic Cyber Security Theory
Skills recruiters demand	Deep Learning (PyTorch/TensorFlow), LLM Fine-Tuning & RAG, MLOps Pipelines, Penetration Testing & Ethical Hacking, ROS2 Robotics, Cloud AI
Missing skills gap	Production MLOps, LLM Architecture & Deployment, Applied Cybersecurity Operations, Real-World DeepTech Prototyping
Emerging tech requirements	Agentic AI Frameworks, Quantum Computing Basics, Autonomous Systems, Zero-Knowledge Cryptography
Recruiter demand	Very High and rapidly growing — premium demand from AI startups, MNC R&D centers, FinTech and DeepTech.
Representative top recruiters	Microsoft AI, Google DeepMind, Nvidia, IBM Research, Fractal Analytics, Palantir, Razorpay

This table documents the branch with the fastest-moving demand profile in the database, even though its overall Skill Gap Score (29.5) is low. The taught row (Python basics, introductory ML theory) is genuinely foundational, while the demanded row (LLM fine-tuning, MLOps pipelines, agentic AI frameworks) reflects technology that did not widely exist when most current curricula were designed. Recruiter demand is 'Very High and rapidly growing,' pulling from AI startups and global R&D centers (Microsoft AI, Google DeepMind, Nvidia) that pay a premium for exactly the missing skills listed. This table is the clearest illustration of the 'velocity gap' pattern referenced in the Key Insights: the shortfall here is about how fast curricula can be refreshed, not about foundational quality.

5.7 Computer Science Engineering (CSE) — Skill Gap Score: 27.8 (Low)

Dimension	Details
Skills currently taught	C/C++, Core Java, Basic Data Structures, RDBMS/SQL, HTML/CSS, Operating System Concepts, Computer Networks Theory
Skills recruiters demand	Advanced Problem Solving, System Design (LLD/HLD), Modern JavaScript/TypeScript, React/Node.js, Docker/Kubernetes, Cloud Infrastructure (AWS/GCP), CI/CD Pipelines, Microservices
Missing skills gap	Production-grade Cloud DevOps, System Scalability, Production Debugging, Agile Workflow, Git Collaboration, Modern Frameworks
Emerging tech requirements	Generative AI/LLM Orchestration, Vector Databases (Pinecone/Chroma), Distributed Computing, Edge AI, MLOps
Recruiter demand	Very High — the top tech-sector demand pool: SaaS, FinTech, E-commerce, Global Capability Centers.
Representative top recruiters	Google, Microsoft, Amazon, Meta, Flipkart, TCS, Infosys, Goldman Sachs

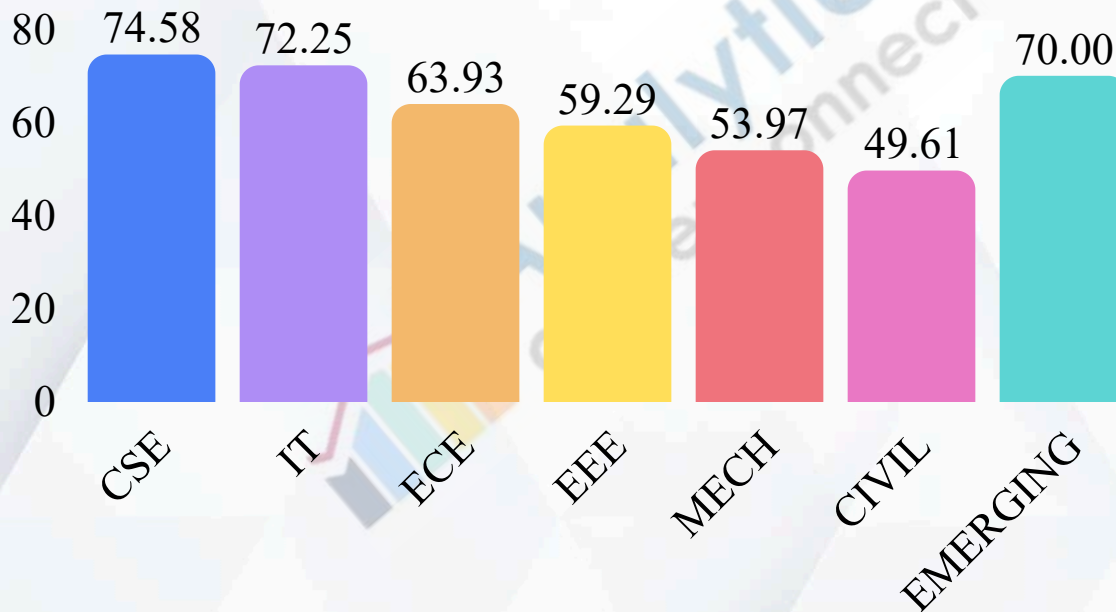
This table represents the benchmark branch, with the lowest Skill Gap Score (27.8) in the dataset. The taught row (C/C++, data structures, RDBMS, OS concepts) forms a strong computer-science foundation, and the demanded row (system design, Docker/Kubernetes, cloud infrastructure, CI/CD) builds directly on it rather than requiring a different skill family. The missing-skills row is concentrated in production engineering practices — DevOps, scalability, debugging at scale — that are typically learned through internships and real projects rather than classroom instruction, which is why this report's recommendations emphasize continuous, project-linked upskilling for this branch rather than curriculum overhaul. Recruiter demand is 'Very High' from the broadest employer base in the report (Google, Microsoft, Amazon, TCS, Goldman Sachs), confirming this branch's position as the intelligence platform's reference standard for industry readiness.

Branch Comparison Analysis

6.1 Placement % By Branch Comparison

Within every branch, government-run institutions (IITs, NITs, IIITs, government colleges) outperform private institutions on placement percentage. The gap is widest in the branches that already have the largest overall skill gap — Civil Engineering (13.8 percentage points) and Mechanical Engineering (13.6 points) — suggesting that private colleges in core branches are the segment most in need of curriculum and industry-linkage support.

Placement % By Branch



Branch	Gov. Placement %	Private Placement %	Gap(Points)
CIVIL	60.5	46.6	13.8
MECH	64.6	51.0	13.6
EEE	69.5	56.5	12.9
ECE	74.0	61.2	12.8
EMERGIN G	79.0	67.6	11.4
CSE	82.7	72.4	10.3
IT	80.0	70.1	9.9

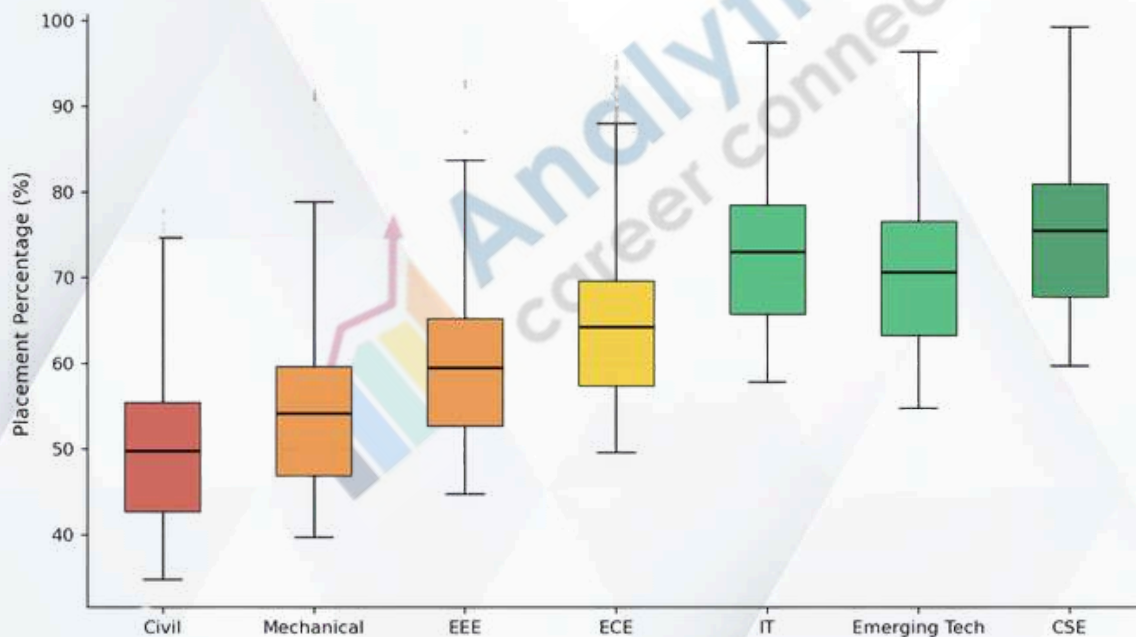
This table isolates the effect of institution ownership from the effect of branch by comparing government and private placement percentages for the same branch side by side. Sorting by gap size (rightmost column) shows the pattern is not random: Civil (13.8 points) and Mechanical (13.6 points) — the two branches with the highest Skill Gap Scores in Table 1 — also have the largest government-private divide, while IT (9.9 points) and CSE (10.3 points) have the smallest. This means the branches that most need curriculum modernization are also the branches where private-college students are most disadvantaged relative to their government-college peers, compounding the risk for that specific student population. The consistent ranking across both tables (branch urgency and ownership gap moving together) is the evidence behind this report's recommendation to prioritize private institutions in Civil and Mechanical Engineering first.

Statistical Deep-Dive & Advanced Analysis

This section goes beyond branch averages to examine consistency, correlation strength, and salary distribution across all 28,091 records, giving ACC a fuller statistical picture of where risk and opportunity are concentrated.

7.1 Consistency of Placement Outcomes (Variability Analysis)

An average placement rate hides how consistent that outcome is across colleges. The box plot below shows the full spread — median, interquartile range, and outliers — of placement percentage within each branch. A wide box means outcomes vary sharply from college to college within that branch; a narrow, high box means most colleges in that branch deliver similarly strong outcomes.



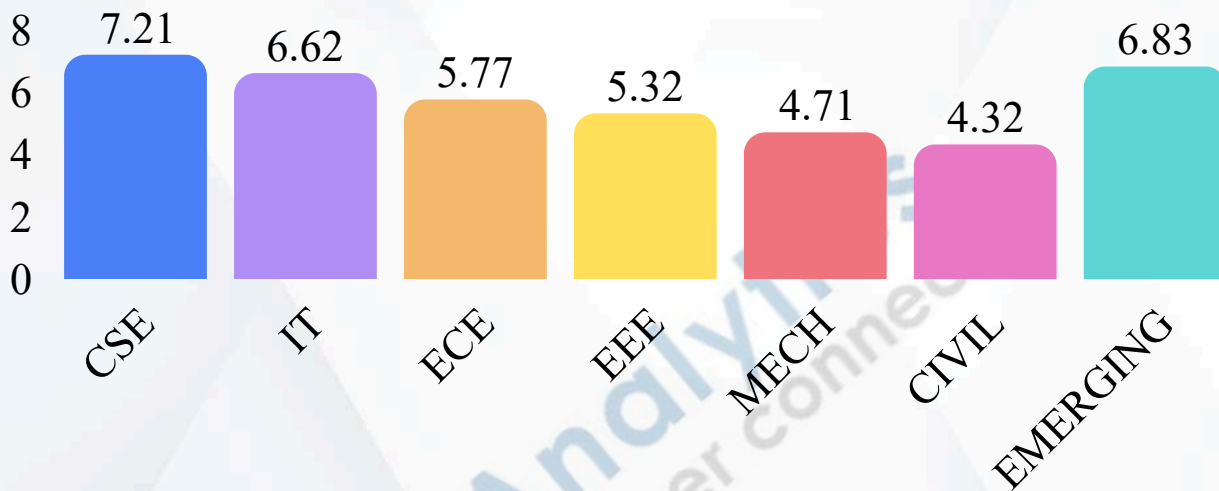
Branch	Mean%	Std. Dev.	Min %	Max %	Range(Max-Mean)
CIVIL	49.6	8.7	34.7	87.5	52.8
MECH	54.0	8.5	39.7	92.0	52.3
EEE	59.3	8.4	44.7	93.7	49.0
ECE	63.9	8.3	49.6	96.0	46.4
EMERGING	70.0	8.3	54.7	98.1	43.4
CSE	74.6	8.0	59.7	99.2	39.5
IT	72.3	7.8	57.8	97.9	40.1

This table adds a risk dimension that the mean alone cannot show. The standard deviation column measures how much placement percentage swings from college to college within the same branch — a higher number means a student's outcome depends more on which specific college they attend, not just which branch they choose. Civil Engineering and Mechanical Engineering carry both the lowest mean placement and among the highest standard deviations (8.7 and 8.5 points respectively), meaning these branches combine a double risk of low average performance and high unpredictability. CSE and IT, by contrast, combine a high mean with a comparatively tighter spread (7.8–8.0 points), indicating the industry-readiness advantage holds fairly consistently across most colleges offering these branches, not just at the top few. The Range column (Max–Min) confirms the same story from a different angle — Civil's 52.8-point range is the widest of any branch, meaning its best and worst colleges are worlds apart, while CSE's 39.5-point range is comparatively the narrowest.

7.2 Salary Distribution Profile

Across all 28,091 records, the average salary distribution is right-skewed: a small number of elite placements (mainly at IITs, in CSE and Emerging Technology) pull the mean above the median, while the bulk of records cluster in a moderate band.

Avg Salary By Branch



- Overall mean salary: ₹5.82 LPA · Median: ₹5.42 LPA · Range: ₹2.86 LPA to ₹30.96 LPA.
- The gap between mean and median (₹0.40 LPA) confirms a modest right-skew driven by top-tier institutions and technology branches, rather than extreme outliers alone.
- The 25th–75th percentile band (₹4.45 LPA – ₹6.76 LPA) represents the realistic outcome range for the majority of graduates, and is the band ACC's upskilling interventions should aim to shift upward for the weaker branches.

Institution- and Region-Level Analysis

8.1 Institution Tier Effect: Does College Category Close the Gap?

Every branch was further split by institution tier — IIT, NIT, IIIT, other Government institutions, State/Other public institutions, and Private colleges — to test whether a strong institution can offset a weak branch, or whether the branch effect dominates regardless of institution quality.

Branch	IIT	NIT	IIIT	Govt.	State/Other	Private	IIT-Private Gap
CIVIL	82.0	70.8	64.6	59.3	55.7	46.6	35.4
MECH	87.9	74.2	69.4	63.5	59.8	51.0	36.9
EEE	88.7	78.5	76.7	68.3	64.6	56.5	32.2
ECE	92.1	84.4	87.2	72.6	70.2	61.1	31.0
IT	93.7	88.6	91.3	79.0	76.2	70.1	23.6
EMERGING	93.8	88.0	90.4	77.7	75.1	67.5	26.3
CSE	95.3	91.0	92.2	81.6	79.5	72.3	23.0

This table splits every branch's placement percentage into six institution tiers, letting ACC see whether a strong college can compensate for a weak branch or whether the branch effect dominates regardless of where a student studies. Reading across any row shows institution tier matters enormously — even in Civil Engineering, IITs place 82.0% of students versus 46.6% at private colleges, a 35.4-point swing purely from institution quality. Reading down the IIT column, however, shows the branch effect never disappears: IIT Civil Engineering (82.0%) still trails IIT CSE (95.3%) and even trails NIT-level CSE (91.0%), so a strong institution raises outcomes within a branch but cannot fully compensate for a structurally weaker branch. The rightmost column (IIT–Private Gap) confirms this compounds with the earlier ownership finding — the gap is widest in Civil (35.4 points) and Mechanical (36.9 points), the same two branches flagged as highest priority throughout this report.

8.2 Regional Variation in Placement Outcomes

States and city-clusters with at least 100 records were ranked by average placement percentage to check whether geography, rather than branch mix, might explain outcome differences.

Top States	Placement %	Colleges
Madhya Pradesh	71.8	217
Punjab	71.5	126
Haryana	70.6	133
Bihar	69.7	413
Uttarakhand	68.8	112
Kerala	68.6	476
Odisha	68.5	217
Delhi	67.3	182

Bottom Regions	Placement %	Colleges
Ghaziabad, Uttar Pradesh	60.8	140
Jaipur, Rajasthan	60.7	308
Navi Mumbai, Maharashtra	60.6	161
Trichy, Tamil Nadu	60.6	133
Namakkal, Tamil Nadu	60.5	119
Coimbatore, Tamil Nadu	60.5	252
Nashik, Maharashtra	60.5	105

The regional spread (60.5%–71.8%, an 11.3-point range) is far narrower than the branch-wise spread (49.6%–74.6%, a 25-point range) or the institution-tier spread (up to 35 points). This is a critical finding for ACC: which branch a student studies and which type of institution they attend matter substantially more than which state or city they study in. Regional policy alone will not close the skill gap — branch-specific and institution-tier-specific interventions will. These two tables rank every state or city-cluster with at least 100 records by average placement percentage, split into the eight strongest and eight weakest regions, with the sample size (n) shown so small, noisy samples aren't mistaken for genuine trends. The headline number is the range: the gap between the best region (Madhya Pradesh, 71.8%) and the weakest (Nashik, Maharashtra, 60.5%) is only 11.3 percentage points. That is far narrower than the branch-wise spread (49.6%–74.6%, a 25-point range) or the institution-tier spread (up to 35 points) documented in Tables 1 and 4. Several of the 'bottom' regions — Jaipur, Coimbatore — have large sample sizes (n=308, n=252), so the weaker showing is a real signal rather than a small-sample artifact. The practical takeaway for ACC is that which branch a student studies and which type of institution they attend matter substantially more than which state or city they study in, so regional policy alone will not close the skill gap — branch-specific and institution-tier-specific interventions will.

8.3 Best- and Weakest-Performing Colleges

To make the gap concrete, the tables below show the top and bottom five colleges by placement percentage for the branch with the narrowest gap (CSE) and the branch with the widest gap (Civil).

Computer Science Engineering (CSE) — Top 5 Colleges

College	State	Placement %	Avg. Salary (LPA)
IIT (BHU) Varanasi	Uttar Pradesh	99.2	₹26.45
IIT Madras	Tamil Nadu	98.5	₹19.85
IIT Palakkad	Kerala	98.3	₹23.75
IIT Hyderabad	Telangana	97.0	₹20.78
IIT Mandi	Himachal Pradesh	97.0	₹25.65

Computer Science Engineering (CSE) — Bottom 5 Colleges

College	State	Placement %	Avg. Salary (LPA)
Shivalik College of Engineering	Himachal Pradesh	59.7	₹5.72
Jagan Institute of Mgmt. Studies Tech. Campus	Delhi	59.7	₹6.84
Medi-Caps University, School of Engg.	Madhya Pradesh	59.7	₹7.83
Ahmedabad	Gujarat	59.7	₹5.41
Agnel Institute of	Goa	59.7	₹6.86

Together, these two tables frame the full range of outcomes within CSE, the branch with the narrowest skill gap in the report. The top five — all IITs — post placement percentages of 97.0% or above and salaries between ₹19.85 LPA and ₹26.45 LPA, representing the ceiling of what the dataset shows is achievable when top-tier institution quality combines with strong curriculum-industry alignment; the tight spread here (97.0%–99.2%, about 2 points) echoes the low-variability finding from Table 3. The bottom five, spread across five different states with no regional clustering, converge on a placement percentage of essentially 59.7% — and that floor is the key number in this section: even the weakest CSE college in the entire 28,091-record database still outperforms the branch-wide average placement percentage of Civil Engineering (49.6%, Table 1) by a full 10 points, and its salaries (₹5.41–₹7.83 LPA) still exceed the Civil Engineering branch average (₹4.32 LPA).

Civil Engineering — Top 5 Colleges

College	State	Placement %	Avg. Salary (LPA)
IIT Kharagpur	West Bengal	87.5	₹15.10
IIT Gandhinagar	Gujarat	86.3	₹10.12
IIT Mandi	Himachal Pradesh	86	₹17.12
IIT Palakkad	Kerala	85.7	₹12.19
IIT Bombay	Maharashtra	85.6	₹14.87

Civil Engineering — Bottom 5 Colleges

College	State	Placement %	Avg. Salary (LPA)
Sri Sai Ram Engineering College	Tamil Nadu	34.9	₹3.85
M.E.S. College of Engineering	Kerala	34.9	₹2.92
Periyar Maniammai Inst. of Science & Tech.	Tamil Nadu	34.9	₹4.25
Rajarambapu Institute of Technology	Other	34.8	₹3.90
Guru Nanak Dev Engineering College	Punjab	34.7	₹3.83

Together, these two tables frame the full range of outcomes within Civil Engineering, the branch with the widest skill gap in the report. The top five — again all IITs — post placement percentages between 85.6% and 87.5% and salaries from ₹10.12 LPA to ₹17.12 LPA; these are meaningfully lower than the equivalent top-five CSE figures (97.0%–99.2% placement, ₹19.85–₹26.45 LPA) even though both tables represent the best possible outcome within their branch at IIT-tier institutions, showing that top-tier institution quality alone cannot erase a structural, branch-level gap — the same point made by Table 4's finding that IIT Civil Engineering still trails IIT and even NIT-level CSE. The bottom five, spanning four different states (Tamil Nadu, Kerala, Punjab, and one unclassified 'Other') with no strong regional pattern, cluster around a 34.7%–34.9% placement percentage and salaries of roughly ₹2.92–₹4.25 LPA, the weakest outcomes recorded anywhere in the database. The resulting ~52.8-point gap between Civil Engineering's best-case (~86.2%) and worst-case (~34.8%) placement percentage is the largest best-to-worst spread of any branch in the report, and is the single clearest data point behind this report.

Key Insights

- Core branches lag on modern tools, not fundamentals: Civil and Mechanical curricula remain strong on theory (thermodynamics, structural analysis, manual drafting) but weak on the software recruiters now require as standard — BIM/Revit/ETABS for Civil, and 3D parametric modeling/FEA/CFD for Mechanical. The gap is tool-adoption, not conceptual understanding.
- Electronics branches (ECE, EEE) sit at the midpoint: Both are pulled up by fast-growing sectors (semiconductors, EVs, 5G/6G, smart grid) but held back by curricula still centered on classical/manual analysis rather than modern EDA tools, embedded firmware, or power-electronics simulation software.
- Technology branches face a different kind of gap: CSE, IT and Emerging Technology already have the highest placement and salary outcomes, but their 'missing skills' are the newest ones — production-grade cloud/DevOps, Kubernetes, LLM deployment, MLOps. Their gap is a velocity gap: curricula cannot update as fast as the technology itself moves, not a foundational skills deficit.
- Recruiter demand and skill readiness move together: Branches with 'Very High' recruiter demand narratives (CSE, IT, Emerging Tech) also post the highest skill-readiness scores, while Civil Engineering has only 'Moderate' recruiter demand and the lowest readiness — demand and readiness reinforce rather than offset each other.
- Ownership amplifies the branch effect: Government institutions in every branch outperform private ones, but the advantage is largest precisely where the branch-level skill gap is largest (Civil, Mechanical) — meaning private colleges in core branches are a compounded risk segment ACC should prioritize.
- Internship exposure is the strongest actionable lever: Internship participation correlates with placement at $r = 0.71$, stronger than salary's own correlation with placement — making structured internship expansion the single highest-leverage, lowest-cost intervention available to ACC across every branch.
- Institution quality helps but cannot fully offset branch weakness: Even at IITs, Civil Engineering placement (82.0%) trails NIT-level CSE (91.0%). Improving institution quality alone will not close the gap — branch-specific curriculum reform is required in parallel.
- Geography is a minor factor compared to branch and institution tier: Regional placement variation spans only 11.3 points (60.5%–71.8%) versus a 25-point branch spread and a 35-point institution-tier spread, so ACC's interventions should be organized around branch and institution tier, not state-by-state programs.

Recommendations

10.1 Branch-specific Upskilling Priorities

Priority	Branch	Recommended Focus
Immediate (Tier 1)	Civil Engineering	BIM/Revit & ETABS certification, Primavera P6 project scheduling, drone surveying, digital site-management tools
Immediate (Tier 1)	Mechanical Engineering	3D CAD (SolidWorks/CATIA), ANSYS FEA/CFD simulation, GD&T, CNC automation, EV vehicle-dynamics modules
Near-term (Tier 2)	Electrical Engineering (EEE)	MATLAB/Simulink, ETAP/PSCAD, PLC/SCADA automation, EV battery management systems
Near-term (Tier 2)	Electronics & Comm. (ECE)	Verilog/SystemVerilog & Cadence EDA, embedded RTOS/firmware, high-speed PCB design
Continuous (Tier 3)	IT	Full-stack (MERN/MEAN), cloud-native architecture, DevOps, automated testing
Continuous (Tier 3)	CSE	System design at scale, Docker/Kubernetes, CI/CD, cloud infrastructure (AWS/GCP)
Continuous (Tier 3)	Emerging Technology	Applied MLOps, LLM fine-tuning/RAG deployment, penetration testing, ROS2 robotics

10.2 Structural Recommendations for ACC

- Tiered intervention model: Deploy intensive, funded upskilling bootcamps for Tier 1 branches (Civil, Mechanical) in partnership with industry tool vendors (Autodesk, Dassault, ANSYS); lighter-touch certification tracks for Tier 2 (EEE, ECE); and continuous micro-credentialing subscriptions for Tier 3 (CSE, IT, Emerging Tech) to keep pace with fast-moving demand.
- Target private colleges in core branches first: Since the government-private placement gap is widest in Civil and Mechanical, ACC's college-support programs should prioritize private institutions offering these branches.
- Curriculum modernization partnerships: Work with colleges to replace manual/classical-only modules (e.g., manual surveying, 2D drafting) with software-based equivalents that mirror the 'Skills Demanded by Recruiters' column identified in this report.
- Recruiter-linked project exposure: Increase internship and live-project participation — internship participation percentage tracks almost exactly with placement percentage across branches, making it a high-leverage, low-cost intervention point.
- Build a living Skill Heatmap Dashboard: Operationalize Figure 3 as a refreshable dashboard so ACC and partner colleges can track Skill Gap Score movement over time as interventions are rolled out.
- Make internships mandatory and structured, first: Given internship participation's $r = 0.71$ correlation with placement — the strongest single driver identified — ACC should treat guaranteed, structured internship placement (not just optional exposure) as the fastest, lowest-cost lever across every branch, especially Civil and Mechanical where participation is currently lowest.
- Pair institution-tier support with branch reform, not instead of it: Because institution quality alone cannot close the branch gap (Section 8.1), ACC's college-support programs should bundle general quality-improvement funding with mandatory branch-specific curriculum modernization milestones, rather than funding either in isolation.

References

- Wheebox, Confederation of Indian Industry (CII), All India Council for Technical Education (AICTE) & Association of Indian Universities (AIU). India Skills Report 2025. 2025. Available at: pwonlyias.com/current-affairs/india-skills-report/
- ETS, CII, AICTE, AIU & Taggd. India Skills Report 2026 — "CS, IT engineers most employable; women beat men for first time in 5 years." Careers360, Nov. 12, 2025. Available at: news.careers360.com
- Business Standard. "Only 10% of India's 1.5 mn engineering graduates to secure jobs this year." Sep. 16, 2024 (citing TeamLease Services). Available at: business-standard.com
- Careers360. "BTech Courses: AI, computer science fuel enrolment boom to 5-year high, but may soon kill jobs, say experts" (AICTE enrolment data by branch, 2024–25). Oct. 24, 2025. Available at: news.careers360.com
- Sakshi Education. "Rising Demand for B.Tech Seats in India: Record Admissions in 2024–25" (AICTE seat-approval and fill-rate data). Aug. 20, 2025. Available at: education.sakshi.com
- CSIS (Center for Strategic and International Studies). "Semiconductor Clusters in the Making: India's Push for Global Competitiveness" (workforce shortfall projections for ISM). 2026. Available at: csis.org
- India Briefing. "Workforce Trends in India's Semiconductor Industry" (Chips-to-Startup and SMART Lab skilling initiatives). Apr. 25, 2025. Available at: india-briefing.com
- NASSCOM. Technology Sector in India: Strategic Review 2026 (in partnership with Indeed) — AI-driven hiring and skill-mix shift. 2026. Available at: community.nasscom.in
- NASSCOM. "India's AI Talent Crisis Is Real — And It's Costing Us the Future" (AI-skilled workforce readiness gap). Feb. 9, 2026. Available at: community.nasscom.in
- NASSCOM & Deloitte India. "India's AI Talent Pool to Grow to 1.25 Million by 2027." IndiaAI.gov.in. Available at: indiaai.gov.in
- NASSCOM. "Cloud Skills: Powering India's Digital DNA" (cloud talent demand-supply gap, 2021–2025). Available at: community.nasscom.in

Conclusion

The Branch-wise Skill Gap analysis shows that India's engineering employability challenge is not evenly distributed — it is concentrated in Civil Engineering and Mechanical Engineering, where curricula remain anchored in classical, manual methods while the industry has moved to modern software and digital tools. Electronics-oriented branches (ECE, EEE) sit at a moderate, improvable gap driven by the same tool-adoption lag. Technology branches (CSE, IT, Emerging Technology) are comparatively industry-ready but face a continuous 'refresh' challenge as AI, cloud-native and MLOps skills evolve faster than syllabi can be revised.

For ACC, this translates into a clear, evidence-based action plan: prioritize funded, tool-focused upskilling in Civil and Mechanical Engineering; run targeted software/tooling certification in EEE and ECE; and support continuous micro-credentialing in CSE, IT and Emerging Technology so these branches retain their competitive edge. Executed together, these branch-specific pathways form the foundation of the Branch-wise Industry Readiness Intelligence Platform ACC set out to build.